

WT Experiment 4

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AIM: Component and Arduino Introduction

Component and Arduino Introduction

The Arduino Uno is a versatile and widely used microcontroller board that acts as the heart of many electronics projects. It is an open-source platform that allows users to design and develop interactive electronic devices. The board consists of an Atmega328 microcontroller that can be programmed to control various hardware components, making it ideal for a variety of applications in IoT, robotics, home automation, and more.

In this experiment, we will explore how to use an Arduino Uno to display real-time temperature readings from a TMP36 temperature sensor on a 16x2 LCD display. The main goal is to read the analog temperature data, convert it into a human-readable temperature format, and display it on the LCD screen.

Arduino Uno Overview

The **Arduino Uno** is one of the most popular boards in the Arduino family due to its simplicity, low cost, and ease of use. It features:

- **Microcontroller:** Atmega328P
- **Operating Voltage:** 5V
- **Digital I/O Pins:** 14 (of which 6 can be used as PWM outputs)
- **Analog Input Pins:** 6
- **Flash Memory:** 32 KB
- **Clock Speed:** 16 MHz
- **USB Connection:** For programming and communication
- **Power Supply:** Can be powered via USB or an external power supply

The Arduino Uno can be programmed using the Arduino IDE, which allows users to write and upload code that will control various electronic components connected to the board.

Components Used in This Experiment

1. Arduino Uno Board:

- Acts as the central controller that processes the sensor data and drives the LCD.

2. TMP36 Temperature Sensor:

- A low-power analog temperature sensor that provides temperature readings in the form of a voltage. This sensor is connected to the Arduino's analog input pin (A0).
- The sensor's output voltage changes linearly with temperature. The TMP36 sensor has a 500 mV offset at 0°C, and its output voltage increases by 20 mV per degree Celsius.

3. 16x2 LCD Display:

- A liquid crystal display used to show the temperature readings. It has 16 characters in each of its 2 rows and can be used to display messages or numbers.
- The `LiquidCrystal` library is used to control the display and print data onto it.

4. Jumper Wires:

- Used to connect the components (Arduino, sensor, and LCD) together on a breadboard or circuit.

5. Breadboard:

- A tool for creating prototype circuits without soldering. It provides a simple way to build the connections between components.

Code Explanation

The following code reads the analog output of the TMP36 temperature sensor, converts it to Celsius, and displays the temperature on the 16x2 LCD display.

```
#include <LiquidCrystal.h>
```

```
// Pin for the temperature sensor
```

```
const int tempPin = A0; // TMP36 sensor connected to A0 pin
```

```
// Initialize LCD
```

```
LiquidCrystal lcd_1(12, 11, 5, 4, 3, 2);
```

```

void setup()

{

  lcd_1.begin(16, 2); // Set up the number of columns and rows on the LCD

  lcd_1.print("Temperature:"); // Print a welcome message on the LCD

}


void loop()

{

  // Read the temperature sensor (analog value between 0 and 1023)

  int sensorValue = analogRead(tempPin);


  // Convert the sensor value to voltage (0-5V range)

  float voltage = sensorValue * (5.0 / 1024.0); // Convert analog value to voltage

  // Convert the voltage to temperature (in Celsius) for TMP36

  // TMP36 has 500mV offset at 0°C, and each degree Celsius increases the voltage by 20mV

  float temperatureC = (voltage - 0.5) / 0.01;

  // Set the cursor to the first row, second column (0,1)

  lcd_1.setCursor(0, 1);

  // Print the temperature value to the LCD

  lcd_1.print(temperatureC); // Display temperature in Celsius

  lcd_1.print(" C  "); // Display unit (Celsius) and some padding


  // Wait for a short time before updating the temperature reading

  delay(2000); // Update every 2 seconds

}

```

Code Breakdown

1. Libraries:

- The `LiquidCrystal.h` library is included to allow easy control of the 16x2 LCD display.

2. Pin Setup:

- The TMP36 temperature sensor is connected to the analog pin A0.

3. LCD Initialization:

- The LCD is initialized with 16 columns and 2 rows. It is used to display messages.

4. Sensor Reading:

- The `analogRead()` function reads the sensor's output, which is an analog voltage between 0 and 1023 (representing 0 to 5V).

5. Temperature Calculation:

- The voltage is calculated from the sensor reading and converted to Celsius using the TMP36 sensor's characteristics: it has a 500 mV offset at 0°C, and the temperature increases by 20 mV for each degree Celsius.

6. Display:

- The calculated temperature is displayed on the second row of the LCD.

7. Loop:

- The process repeats every 2 seconds, updating the temperature display.

Arduino Installation

The screenshot shows the Arduino IDE 2.3.4 download page. The header includes the Arduino logo, navigation links for Professionals, Education, and Makers, and a top bar with Products, Community, Documentation, a SHOP button, a search icon, and a Cloud icon. The main heading is "Downloads". The content area features the Arduino IDE 2.3.4 logo and title, followed by a description of the new major release. It includes links to documentation and source code, and mentions nightly builds. A teal sidebar on the right lists download options for Windows (Win 10 and newer, 64 bits), Linux (AppImage 64 bits (X86-64), ZIP file 64 bits (X86-64)), and macOS (Intel, 10.15: "Catalina" or newer, 64 bits; Apple Silicon, 11: "Big Sur" or newer, 64 bits), along with a link to Release Notes.

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Downloads



Arduino IDE 2.3.4

The new major release of the Arduino IDE is faster and even more powerful! In addition to a more modern editor and a more responsive interface it features autocompletion, code navigation, and even a live debugger.

For more details, please refer to the [Arduino IDE 2.0 documentation](#).

Nightly builds with the latest bugfixes are available through the section below.

SOURCE CODE

The Arduino IDE 2.0 is open source and its source code is hosted on [GitHub](#).

DOWNLOAD OPTIONS

Windows Win 10 and newer, 64 bits

Windows MSI installer

Windows ZIP file

Linux AppImage 64 bits (X86-64)

Linux ZIP file 64 bits (X86-64)

macOS Intel, 10.15: "Catalina" or newer, 64 bits

macOS Apple Silicon, 11: "Big Sur" or newer, 64 bits

[Release Notes](#)

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Download Arduino IDE & support its progress

Since the 1.x release in March 2015, the Arduino IDE has been downloaded **94,270,778** times — impressive! Help its development with a donation.

\$3

\$5

\$10

\$25

\$50

Other

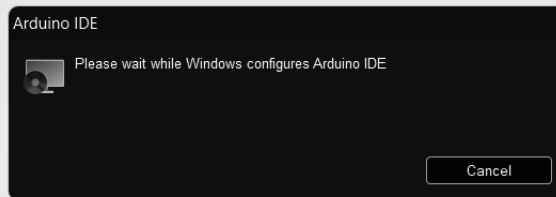
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Name	Date modified	Type	Size
Today			
 arduino-ide_2.3.4_Windows_64bit	26-03-2025 00:13	Windows Installer ...	1,56,120 KB
Yesterday			
 pastry	25-03-2025 10:31	WEBP File	23 KB
 muffins	25-03-2025 10:31	WEBP File	20 KB
 donuts	25-03-2025 10:31	WEBP File	56 KB
 logo			
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 netflix_movies_detailed_up_to_2025			
 AIDS Report	19-03-2025 21:05	Microsoft Word D...	109 KB
 JPMC_CoverLetter	19-03-2025 13:21	Chrome HTML Do...	78 KB
 Mahvich Resume	19-03-2025 12:27	Chrome HTML Do	52 KB



sketch_mar26a | Arduino IDE 2.3.4

File Edit Sketch Tools Help



Select Board



sketch_mar26a.ino

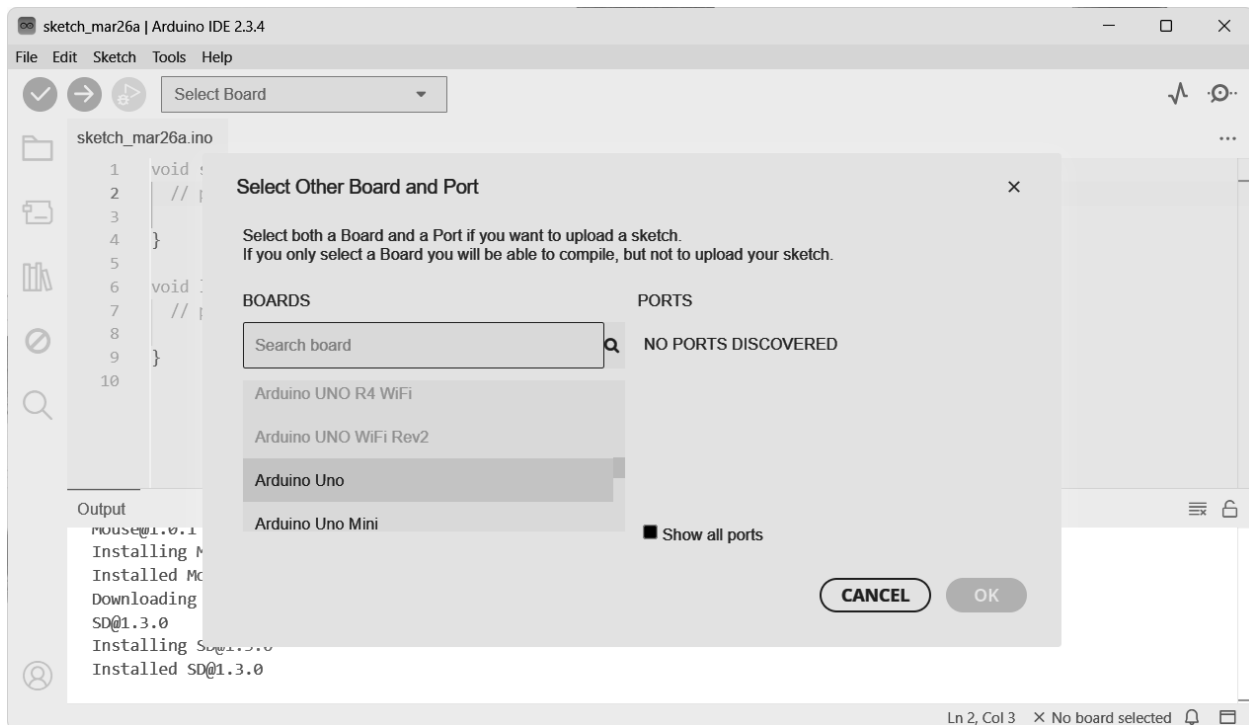
```
1 void setup() {
2   // put your setup code here, to run once:
3
4 }
5
6 void loop() {
7   // put your main code here, to run repeatedly:
8
9 }
10
```

Output

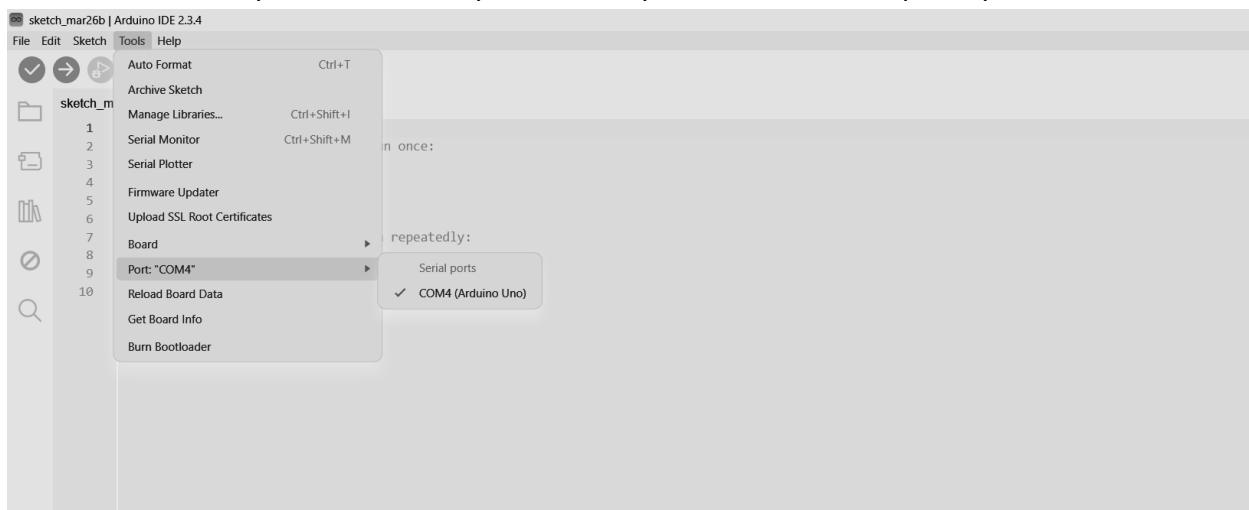
mouse@1.0.1
Installing Mouse@1.0.1
Installed Mouse@1.0.1
Downloading SD@1.3.0
SD@1.3.0
Installing SD@1.3.0
Installed SD@1.3.0

Ln 1, Col 1 X No board selected

Select the required arduino board from select board dropdown list.



In the tools section present on the top bar, select port to select the required port.



Conclusion

This experiment demonstrates how to use the Arduino Uno to interface with a temperature sensor and an LCD display. By combining the Arduino's processing capabilities with simple hardware components like the TMP36 temperature sensor and the LCD, you can easily create a real-time temperature monitoring system. This is a fundamental experiment for understanding analog-to-digital conversion, sensor interfacing, and display output on embedded systems.

